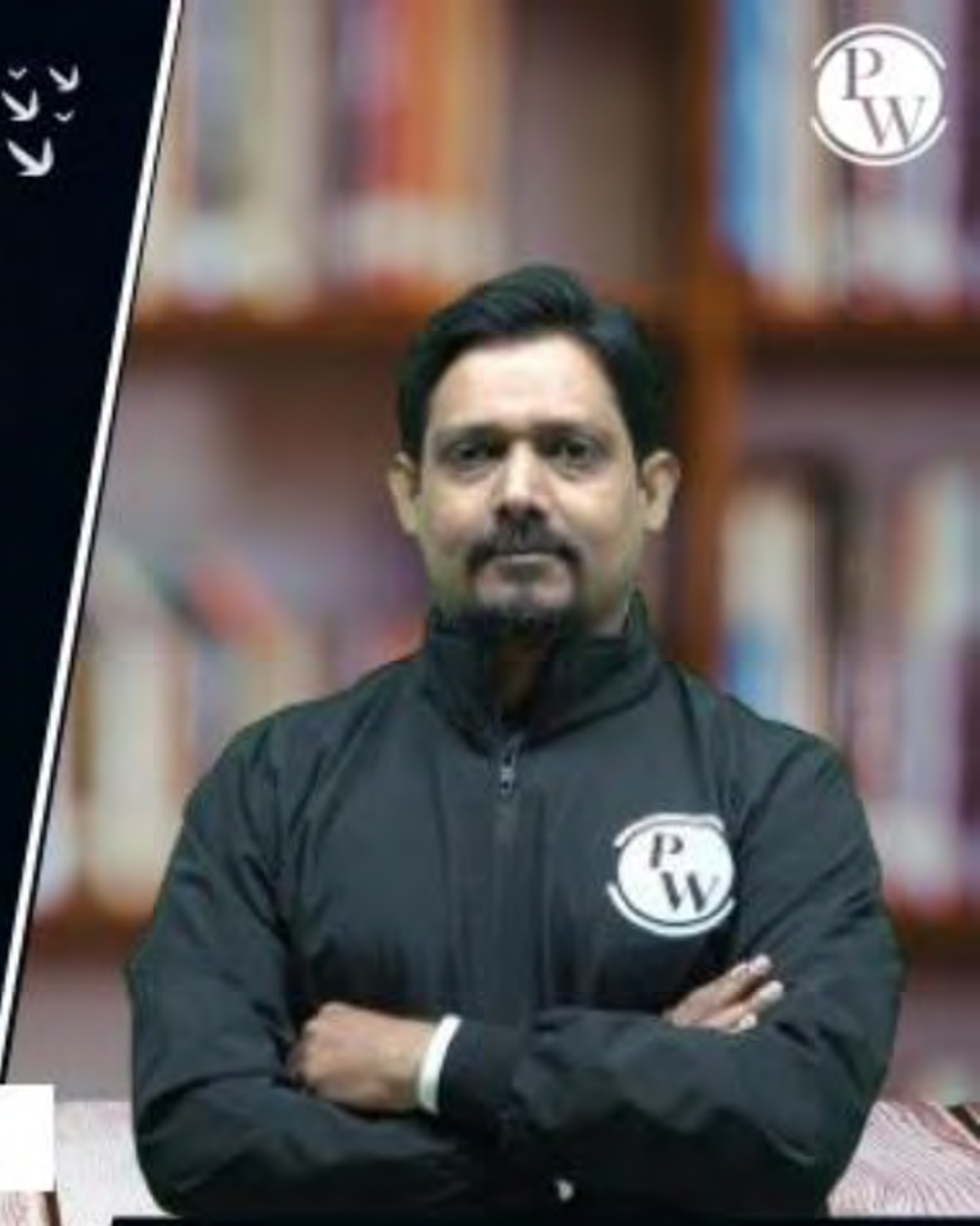
Computer Science & IT



Data Structure & Programming

Linked List

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



- Topic — pointer
- Topic — Structure
- Topic — Dynamic Memory Allocation
- Topic — getnode, build123(), display
- Topic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Single linked list operation

Caching Insert delete.

Slide

Test purpose

1 Revision ✓

2 Time management ✓

3 Recall ✓

4 Identification of weak concept

5. Concept learning

6 Silly Mistake Identification

(8) How to duck Bouncer

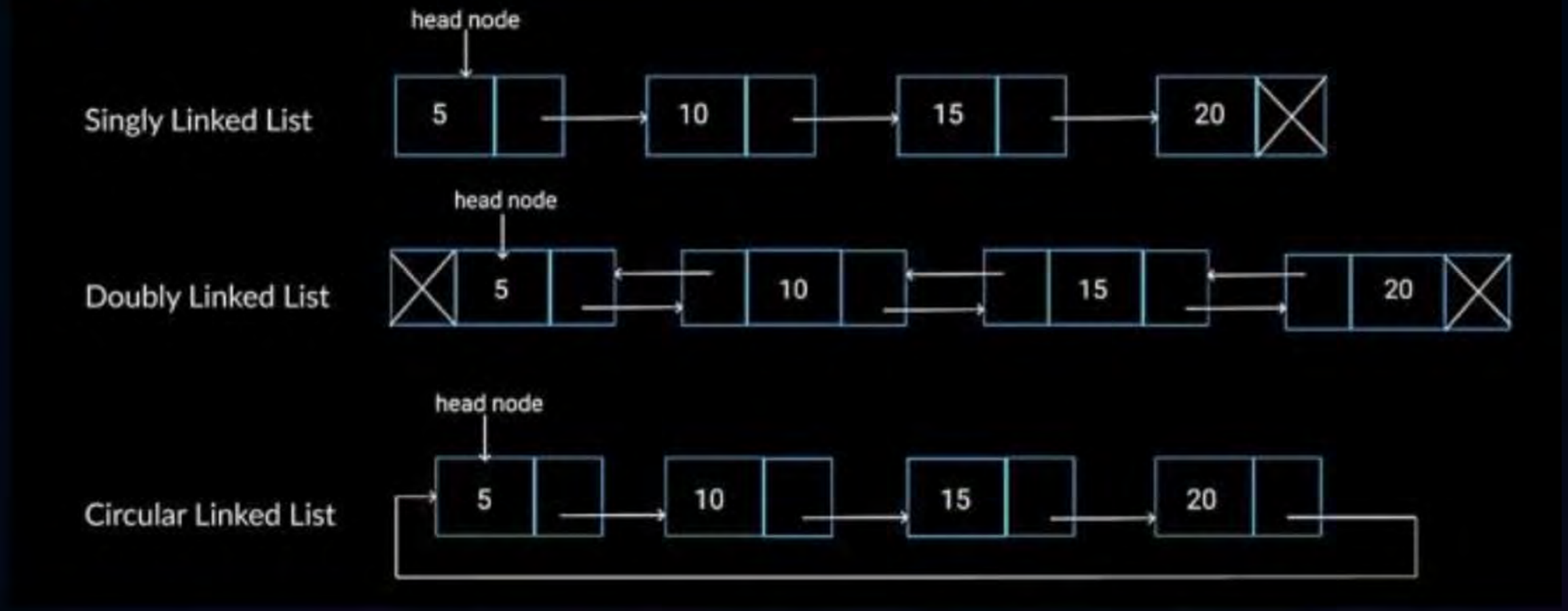
1 year

7/23

7 How to solve Approach



Topic : Linked List



Topic : Single Linked List




```
void display_R(node* HP) {
    if(HP) {
        printf("%d", HP->data);
        display_R(HP->next);
    }
}
```

HP - global variable

HP = 1000

Slide

```
typedef struct node {
        int data;
        struct node* next;
} node;
```

Topic : Single Linked List



if(HP != NULL)

if(HP)

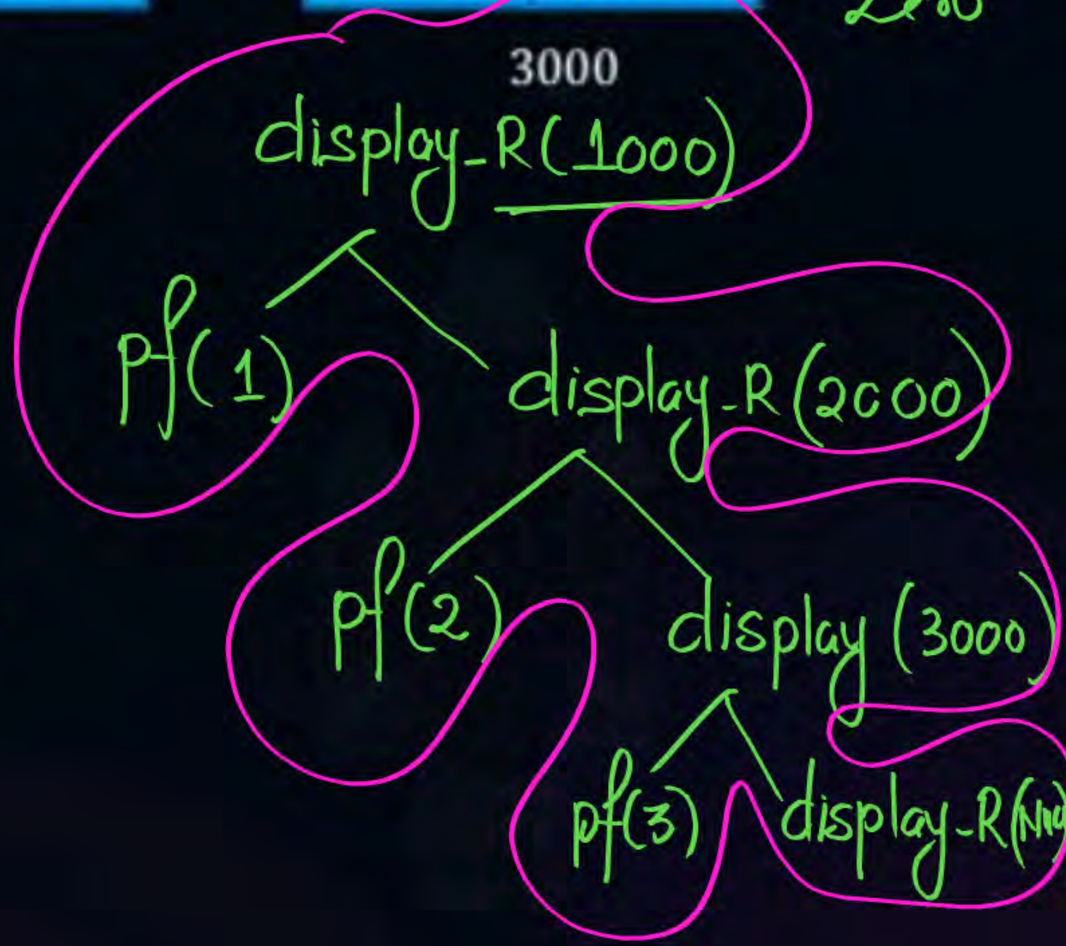
Nonzero True

Zero

Recursive function

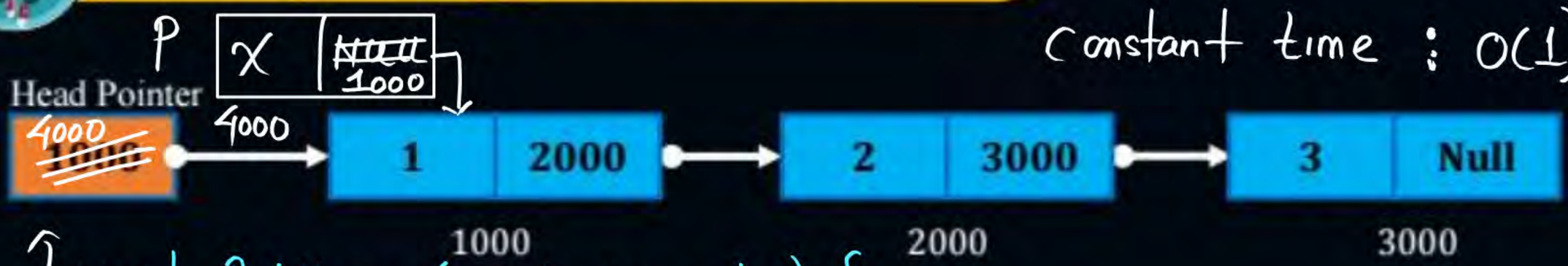
```
void display_R(node* HP) {
    if(HP) {
        printf("%d", HP->data);
        display_R(HP->next);
    }
}
```

1,2,3



Slide

Topic : Single Linked List



Constant time : O(1)

```
void Addbegin( node*s, int x) {
    node * p = getnode(x);
    p→next = S;
    HP = P;
}
```

update to 4000

HP

HP | 1000

No traversal required

Independent of length of list

O(1)

Slide



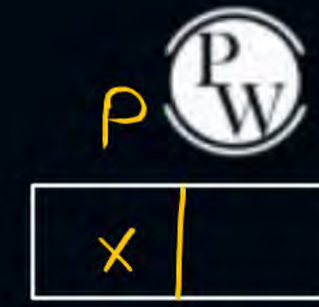
Topic : Single Linked List

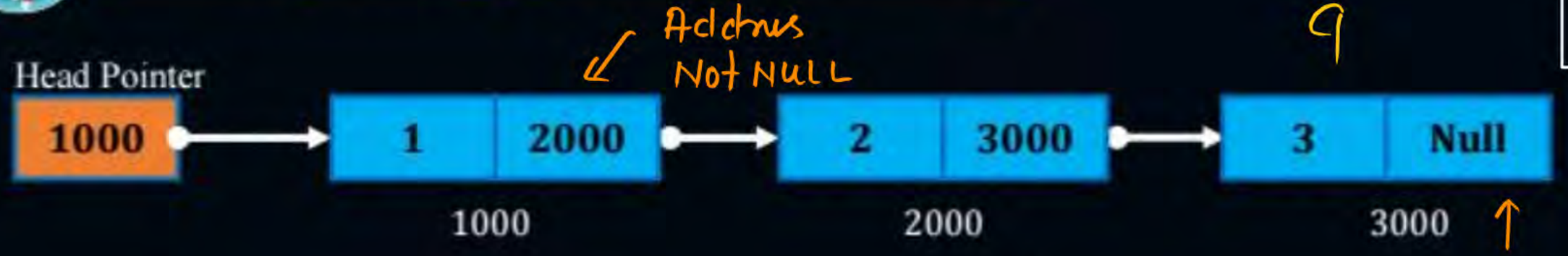
NULL ASCII value zero

Slide



Topic : Single Linked List





Adding a Node at the end of list

1 traverse the list

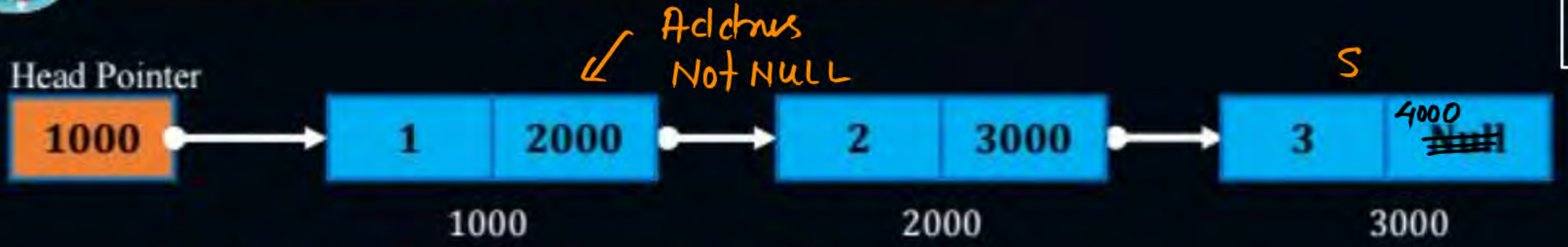
2. we Need to stop at the Last Node

for the Last Node

next part will be NULL

Slide

Topic : Single Linked List



P

x	NULL

4000

1

```
void Addend(node*S, intx){

    node* p = getnode(x);

    while(S→next!=NULL)
        S=S→next;

    S→next = p;

    }
```

Nonempty list

Slide



Topic : Single Linked List

→ dereference (*)

1000 → data 1000 → next

```
node* p;

p=NULL;

printf("%d", p->data);
```

Error

NULL pointer dereference

Slide

Topic : Single Linked List



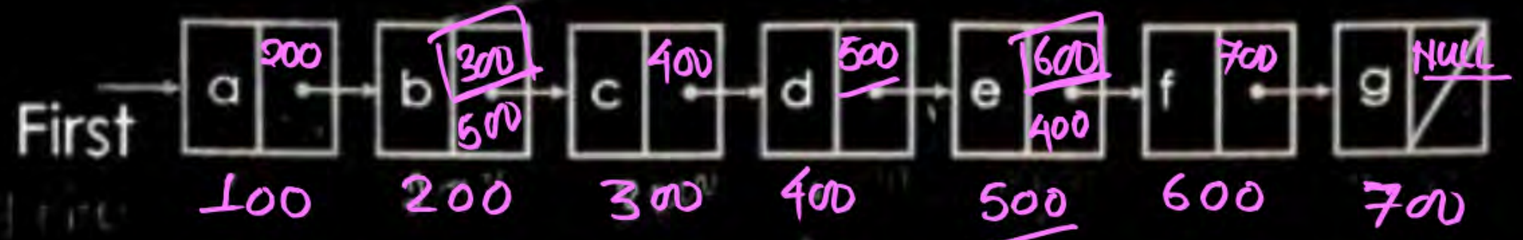

```
void Addend (node *s, int x) {
    node *p = getnode(x);
    if(s) {
        while(s->next != NULL)
            s = s->next;
        s->next = p;
    }
    else
        HP = p;
}
```

```
int main() {
    node *HP; ✓
    HP = Build123()
    addbegin(HP, 4);
    addend(HP, 5);
```

Slide

Topic : Single Linked List

01.



What would be the output after the following sequence of steps?

```
struct node *p;
p = first → link → link → link;
p → link → link = p;
first → link → link = p → link;
printf ("%c", first → link → link → link → data);
```

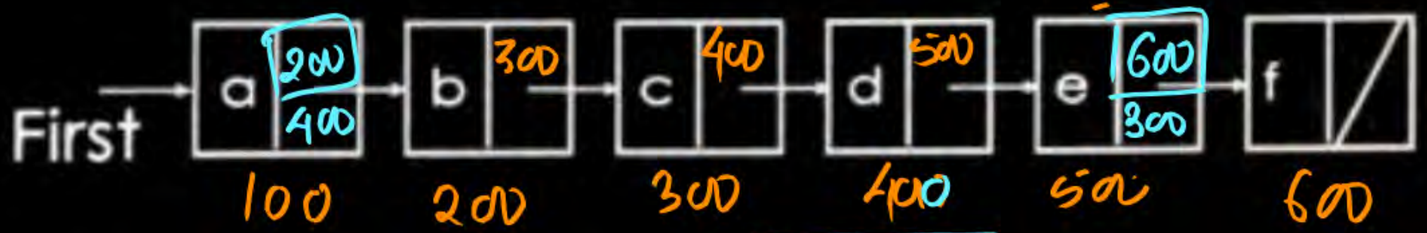
p = 400

(a) a (b) b (c) c (d) d

Slide

Topic : Single Linked List

02. What is the output after the following sequence of steps?



P = 300

```
struct node *P;
(i)  P = first → link → link;
(ii) P → link → link → link = P;
(iii) first →link = P → link;
(iv) printf ("%c", first → link → link →link → link → data);
```

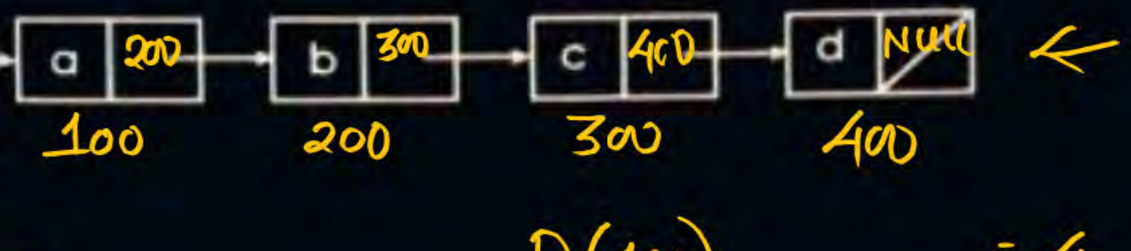
(a) a (b) b (c) c (d) d

Slide

Topic : Question

What does Do( ) do on a singly linked list

Recursive counting program




```
int Do (struct node *p)
{
        if (p)
            return 1 + Do (p → link);
        else
            return 0;
}
void main ( )
{
        printf ("%d", Do (first));
}
```

D(100) = 4

1 + Do(200)

1 + Do(300)

1 + Do(400)

1 + Do(NULL)

0

3

2

1

Slide

(a) 1 (b) 2 (c) 3 (d) 4



Topic : Question

Consider the following function

```
int dosmothing ( struct node * q ){
int c = q -> data ;
if (q==null) return 0;
if (q -> link == NULL) return c;
else
c = c + dosmothing (q = q -> link) ;
return c ;
}
```

The above function run on the linked list that contain 1, 2, 3, 4, 5, 6, 7, 8. What will the content of the linked lists & the return value?

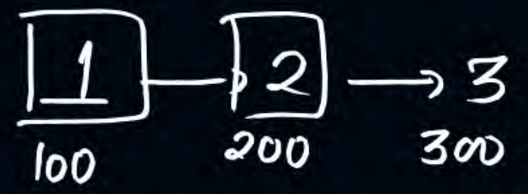
(A) 1, 2, 3, 4, 5, 6, 7, 8 return value 28

(B) 2, 1, 4, 3, 6, 5, 8, 7 return value 36

(C) 1, 2, 3, 4, 5, 6, 7, 8 return value 36

(D) 2, 1, 4, 3, 6, 5, 8, 7 return value 28

Topic Question



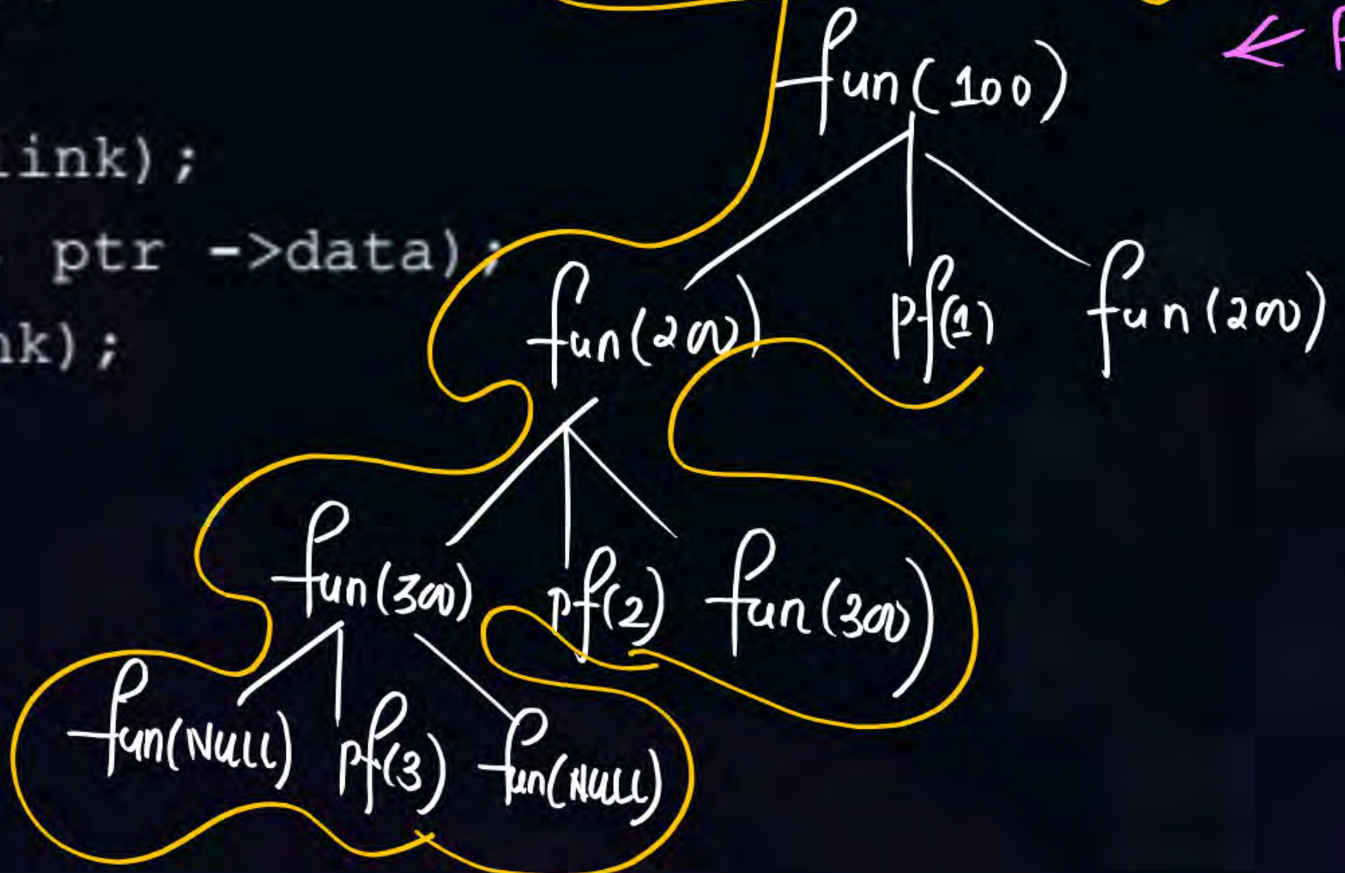
If the list contain the value 1→2→3 then what is the Output following function.

```
void fun (Node *ptr){
if (ptr){
        fun (ptr -> link);
        printf ("%d", ptr ->data);
        fun (ptr->link);
        return;
}
```

fun

No. of values printed by the program?

← partial Recursion



3 2 3 1 3 2 3

Slide



Topic Question

Consider the function defined below.

```
struct item {
        int data;
        struct item * next;
};
int f(struct item *p) {
    return ((p == NULL) || (p->next == NULL)||
        ((p->data <= p ->next -> data) &&
        f(p->next)));
}
```

For a given linked list p , the function f returns 1 if and only if

A the list is empty or has exactly one element

B the elements in the list are sorted in non-decreasing order of data value

C the elements in the list are sorted in non-increasing order of data value

D . not all elements in the list have the same data value

Slide



Topic Question

1 1 22,333

Consider the function defined below.

```
struct item {
        int data;
        struct item * next;
};
int f(struct item *p) {
    return ((p == NULL) || (p->next == NULL)||
        ((p->data <= p ->next -> data) &&
        f(p->next)));
}
```

p→data <= p→next→data

1 2 3 4 5 6 Increasing

1 <= 2 <= 3 f( )

2 <= 3

Slide



2 mins Summary

Topic — Addbegin

Topic — Addend

Topic — problem

Topic

Topic

Slide

THANK - YOU